

2 Ratio Test

The Ratio Test

To test the series

$$\sum_{n=1}^{\infty} a_n,$$

take the limit

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L = \lim_{n \rightarrow \infty} |a_{n+1}| \cdot \frac{1}{|a_n|}$$

- If $L < 1$ the series is absolutely convergent.
- If $L > 1$ the series is divergent. (This include if $L = \infty$.)
- If $L = 1$ the ratio test gives no information.

On the remaining examples, determine if the given series is Absolutely Convergent, Conditionally Convergent, or Divergent.

2.0.1 Example:

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n + 4}$$

$$a_n = \frac{n^2}{3^n + 4}$$

$$a_{n+1} = \frac{(n+1)^2}{3^{n+1} + 4}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^2}{3^{n+1} + 4} \cdot \frac{3^n + 4}{n^2} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{n^2} \cdot \frac{3^n + 4}{3^{n+1} + 4}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 + 2n + 1}{n^2} \cdot \lim_{n \rightarrow \infty} \frac{3^n + 4}{3^{n+1} + 4}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2}{n^2} \cdot \lim_{n \rightarrow \infty} \frac{3^n}{3^{n+1}}$$

$$= 1 \cdot \frac{1}{3} = \frac{1}{3} < 1 \quad \therefore \boxed{\text{Converge}}$$

Ex: Use the ratio test on

$$a_{n+1} \quad \frac{1}{a_n}$$

$$\sum_{n=0}^{\infty} \frac{2^{n+1}}{5^n}$$

$$\lim_{n \rightarrow \infty} \frac{2^{n+2}}{5^{n+1}} \cdot \frac{5^n}{2^{n+1}} =$$

$$= \lim_{n \rightarrow \infty} \frac{2^{n+2}}{2^{n+1}} \cdot \frac{5^n}{5^{n+1}} = \lim_{n \rightarrow \infty} \frac{\cancel{2^n} \cdot 2^2}{\cancel{2^n} \cdot 2^1} \cdot \frac{\cancel{5^n}}{\cancel{5^n} \cdot 5^1} = \frac{\cancel{2} \cdot 2}{\cancel{2} \cdot 5} = \frac{2}{5}$$

$$= \frac{2}{5} < 1 \quad \therefore \boxed{\text{Converge}}$$

2.0.2 Example:

$$\sum_{n=1}^{\infty} \frac{n^9 + n^2}{n!}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^9 + (n+1)^2}{(n+1)!} \cdot \frac{n!}{n^9 + n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)^9 + (n+1)^2}{n^9 + n^2} \cdot \frac{n!}{(n+1)!}$$

$$= \lim_{n \rightarrow \infty} \frac{\boxed{n^9} + \dots + n^2 + 2n + 1}{\boxed{n^9} + n^2} \cdot \frac{\cancel{n!}}{(n+1)\cancel{n!}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^9}{n^9} \cdot \lim_{n \rightarrow \infty} \frac{1}{n+1} = 1 \cdot 0 = 0 < 1 \quad \therefore \boxed{AC}$$

2.0.3 Example:

$$\sum_{n=1}^{\infty} \frac{n^2 + n + 1}{2^n}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^2 + (n+1) + 1}{2^{n+1}} \cdot \frac{2^n}{n^2 + n + 1} = \lim_{n \rightarrow \infty} \frac{\boxed{n^2} + 2n + 1 + n + 1 + 1}{\boxed{n^2} + n + 1} \cdot \frac{\cancel{2^n}}{\cancel{2^n} \cdot 2}$$

$$= 1 \cdot \frac{1}{2} = \frac{1}{2} < 1$$

$\therefore$ Converges

2.0.4 Example:

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{3n+4}{n^2} \right)$$

$$\lim_{n \rightarrow \infty} \frac{3(n+1)+4}{(n+1)^2} \cdot \frac{n^2}{3n+4} = \lim_{n \rightarrow \infty} \frac{3n+3+4}{3n+4} \cdot \frac{n^2}{n^2+2n+1}$$

$1 \cdot 1 = 1$ inconclusive

→ AST.

$$\lim_{n \rightarrow \infty} \frac{3n+4}{n^2} = 0 \therefore C. \text{ by AST.}$$

2.0.5 Example:

AST $\lim_{n \rightarrow \infty} \frac{3^n}{n^2 \cdot 2^n} = ?$

$$\sum_{n=1}^{\infty} (-1)^n \left(\frac{3^n}{n^2 \cdot 2^n} \right)$$

Ratio Test $\lim_{n \rightarrow \infty} \frac{3^{n+1}}{(n+1)^2 2^{n+1}} \cdot \frac{n^2 \cdot 2^n}{3^n}$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{3} \cdot 3}{\cancel{3}^n} \cdot \frac{n^2}{n^2+2n+1} \cdot \frac{\cancel{2}^n}{\cancel{2}^n \cdot 2}$$

$$= 3 \cdot 1 \cdot \frac{1}{2} = \frac{3}{2} > 1$$

Diverges

2.0.6 Example:

$$\sum_{n=1}^{\infty} \frac{(-1)^n n^3 \cdot 10^n}{n!} \quad \lim_{n \rightarrow \infty} \frac{(n+1)^3 \cdot 10^{n+1}}{(n+1)!} \cdot \frac{n!}{n^3 \cdot 10^n}$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)^3}{n^3} \cdot \frac{10^{n+1}}{10^n} \cdot \frac{n!}{(n+1)n!}$$

$$= \lim_{n \rightarrow \infty} \frac{n^3 + 3n^2 + 3n + 1}{n^3} \cdot \frac{10^n \cdot 10}{10^n} \cdot \frac{n!}{(n+1)n!}$$

$$1 \cdot 10 \cdot 0 = 0 < 1$$

$\therefore$
Converge

1. $\sum \frac{n^2}{e^n}$

Series Practice Worksheet 6

The Ratio Test

Use the ratio test to determine if each series is convergent or divergent.

$$1. \sum_{n=1}^{\infty} n^2 e^{-n} \quad \lim_{n \rightarrow \infty} \frac{(n+1)^2}{e^{n+1}} \cdot \frac{e^n}{n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 + 2n + 1}{n^2} \cdot \frac{\cancel{e^n}}{\cancel{e^n} \cdot e} = \frac{1}{e} < 1 \therefore C$$

$$2. \sum_{n=1}^{\infty} \frac{n^2 2^{n-1}}{(-5)^n} \quad \lim_{n \rightarrow \infty} \frac{(n+1)^2 2^n}{5^{n+1}} \cdot \frac{5^n}{n^2 2^{n-1}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 + 2n + 1}{n^2} \cdot \frac{\cancel{5^n}}{\cancel{5^n} \cdot 5} \cdot \frac{2^n}{2^n \cdot 2^{-1}} = 1 \cdot \frac{1}{5} \cdot 2 = \frac{2}{5} < 1 \quad \boxed{C}$$

$$3. \sum_{k=1}^{\infty} \frac{2^k k!}{(k+2)!} \quad \lim_{k \rightarrow \infty} \frac{2^{k+1} (k+1)!}{(k+3)!} \cdot \frac{(k+2)!}{2^k k!}$$

$$= \lim_{k \rightarrow \infty} \frac{2^k \cdot 2}{2^k} \cdot \frac{(k+1) \cancel{k!}}{\cancel{k!}} \cdot \frac{(k+2)!}{(k+3)(k+2)\cancel{k!}} = \lim_{k \rightarrow \infty} 2 \cdot \frac{k+1}{k+3}$$

$$= 2 > 1 \quad \mathcal{D}$$

$$4. \sum_{n=1}^{\infty} \frac{2^{n^2}}{n!} \quad \lim_{n \rightarrow \infty} \frac{2^{(n+1)^2}}{(n+1)!} \cdot \frac{n!}{2^{n^2}} = \lim_{n \rightarrow \infty} \frac{2^{n^2+2n+1}}{2^{n^2}} \cdot \frac{\cancel{n!}}{(n+1)\cancel{n!}}$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{2^{n^2}} \cdot 2^n \cdot 2^1}{\cancel{2^{n^2}}} \cdot \frac{1}{n+1} = \lim_{n \rightarrow \infty} \frac{2^{2n} \cdot 2}{n+1} = \infty$$

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